2. Decoding the video stream and displaying a series of images on the screen
3. Decoding the audio stream and sending the sound to your speakers

The word “codec” is a portmanteau, a combination of the words “coder” and “decoder.” Audio and video codecs are just algorithms used to encode and decode a particular audio or video stream, in a particular way. Raw media files can be enormous. Without encoding a video or audio clip, it would be too large to transmit across the Internet in a reasonable amount of time. Similarly without a decoder, the recipient would not be able to reconstitute the original media source from the encoded form. A codec is able to understand a specific container format and decodes the audio and video tracks that it contains.

A codec can be either lossy or lossless. A lossy video codec means that information is being irretrievably lost during encoding while lossless codec allows the exact original data to be reconstructed from the compressed data. Lossless codecs are much too big to be useful on the web and so only lossy codecs are used.

Common video codecs

H.264 - It is also known as “MPEG-4 part 10,” a.k.a. “MPEG-4 AVC,” a.k.a. “MPEG-4 Advanced Video Coding.” It aims to provide a single codec for low-bandwidth, low-CPU devices (cell phones); high-bandwidth, high-CPU devices (modern desktop computers); and everything in between.

THEORA – This is a royalty-free codec and is not encumbered by any known patents other than the VP3 patents, from which it evolved originally. Theora video can be embedded in any container format, although it is most often seen in an Ogg container.

VP8 – This is another codec from On2, the same company that originally developed VP3 (later Theora). Technically, it produces output on par with H.264 High Profile, while maintaining a low decoding complexity on par with H.264 Baseline.

The most common audio codecs are:

- MPEG-1 Audio Layer 3 – You may know them as MP3. They can be encoded at different bitrates like 64 kbps, 128 kbps, 192 kbps, and so on. Higher bitrates mean larger file sizes and better quality audio. The MP3 format allows for variable bitrate encoding, which means that some parts of the encoded stream are compressed more than others.
- Advanced Audio Coding – AAC was designed to provide better sound quality than MP3 at the same bitrate, encode audio at any bitrate and up to 48 channels of sound. Like H.264 it also defines multiple profiles and it is patent encumbered.